PROJECT EXHIBITION 01

Breast Cancer Detection Using Histopathology and Machine Learning

Project Report

A reproducible, explainable CNN-based prototype with an image validation gate and Streamlit interface

Project Type	AI/ML educational and research prototype
Primary Task	Breast histopathology image classification
Output	Benign / Malignant + model confidence
Primary Model	3-Stage CNN
Evaluation Set	1,664 unseen test images; zero patient leakage
Overall 3-Class Accuracy	81.61%

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1. Executive Summary

Project Exhibition 01 is a focused AI/ML prototype for breast histopathology image classification. The system accepts an uploaded image, performs a lightweight validation and segregation step before model inference, preprocesses accepted images, and uses a single convolutional neural network to classify the sample as Benign or Malignant. A Streamlit interface presents the prediction and model confidence to the user.

The supplied evaluation materials report performance on 1,664 unseen test images under a zero-patient-leakage evaluation setup. The computed three-class result across Benign, Invalid, and Malignant inputs is 81.61% accuracy, with 81.03% weighted precision, 81.61% weighted recall, and 81.07% weighted F1-score. Malignant recall is 90.30%, while the input gate rejects 69 of 75 invalid samples, giving 92.00% invalid-class recall.

The project is intentionally designed to remain understandable and reproducible rather than over-engineered. The application does not attempt multimodal cancer prediction, does not use user-editable decision thresholds, and does not place benchmark comparison logic inside the user interface.

2. Problem Statement

Breast cancer histopathology images contain visual cellular and tissue patterns that can be difficult to interpret consistently and quickly. This project explores how a compact deep-learning pipeline can learn from labelled breast histopathology images and provide a binary classification result through a simple software interface.

The project is not intended to replace a pathologist or provide a clinical diagnosis. Its purpose is to demonstrate the complete AI/ML workflow - from data preparation and controlled inference to evaluation and application delivery - in a form suitable for a student exhibition.

3. Objectives

- Build a reproducible histopathology image-classification pipeline.
- Validate uploaded images before running cancer classification.
- Train and save a single CNN model for Benign/Malignant prediction.
- Evaluate the final system on unseen patients without patient-level leakage.
- Measure overall, class-wise, magnification-level, subtype-level, and invalid-input performance.
- Expose the trained model through a simple Streamlit application.
- Keep the implementation understandable enough for live demonstration and viva explanation.

4. Scope and Design Decisions

Area	Decision
Input	Breast histopathology image only
Validation	Lightweight image validation / segregation gate before inference
Prediction	Single CNN; Benign vs Malignant
UI	Streamlit
Backend	No separate Java/FastAPI/backend service in the initial system
Threshold	No user-editable malignancy threshold
Benchmarks	Maintained as report material rather than application functionality
Medical positioning	Educational/research prototype; not a clinical diagnosis

5. System Architecture

The final design uses a linear inference path. The input gate is deliberately placed before the classifier so that unrelated or unusable images are stopped before cancer prediction. This separation is important both technically and from a demonstration perspective.

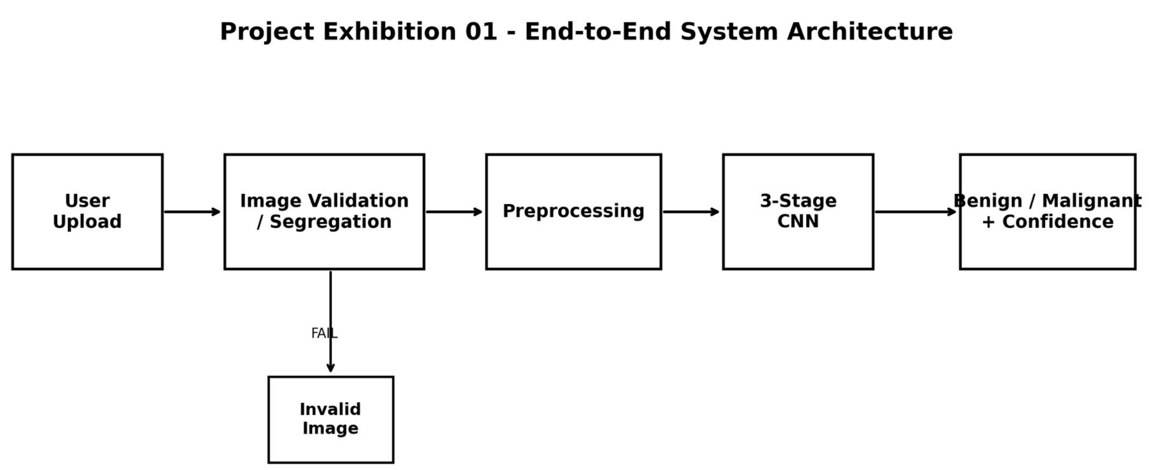

Figure 1. Final end-to-end architecture for Project Exhibition 01.

5.1 Main Components

- Streamlit application - handles upload, preview, validation feedback, prediction result, confidence display, and disclaimer.
- Image validation / segregation - checks whether the upload is usable and suitable enough to enter the cancer-classification path.
- Preprocessing - converts the accepted image to the expected color representation, size, and normalized numerical range.

- Single CNN - performs the Benign/Malignant classification.
- Result layer - displays prediction, model confidence, and a clear non-clinical disclaimer.

6. End-to-End Workflow

1. User selects an image in the Streamlit interface.
2. The system checks that the file is a readable image and performs the implemented suitability checks.
3. If validation fails, the system stops and returns an invalid-image message.
4. If validation succeeds, the image is preprocessed using the same basic steps used for model input.
5. The saved CNN receives the processed image.
6. The model produces a Benign or Malignant prediction with an associated confidence value.
7. The application displays the result and the educational/research disclaimer.

7. Dataset and Data Preparation

The primary dataset used by the project is BreaKHis. The project documentation records 7,909 breast histopathology images from 82 patients, with Benign and Malignant labels and four magnification levels: 40X, 100X, 200X, and 400X. Standard images are recorded as RGB with a 700 x 460 format in the dataset documentation.

A central methodological requirement is patient-level separation. Because multiple images can be associated with the same patient or slide, related images must not be distributed across training, validation, and test sets in a way that causes information leakage. The supplied evaluation report identifies the final test set as unseen-patient evaluation with zero patient leakage.

7.1 Dataset Preparation Checklist

- Verify Benign/Malignant labels.
- Extract patient/slide information from filenames where required.
- Create patient-level train/validation/test splits.
- Inspect class counts and representative images.
- Keep validation and test data free of training-only augmentation.
- Keep raw and processed data logically separated.

8. Image Preprocessing

The project uses a single preprocessing path so that the representation seen during training matches the representation used during application inference. The documented flow is: open image -> convert to RGB -> resize -> scale pixel values.

Step	Purpose
Open image	Confirm the file can be read and obtain image data.
RGB conversion	Standardize the channel representation.
Resize	Match the input dimensions expected by the CNN.
Normalization	Scale pixel values to the numerical range used by the model.

The design intentionally avoids multiple preprocessing pipelines because duplicated or inconsistent preprocessing is a common source of inference errors.

9. CNN Model

The trained model used for the evaluation report is identified as a 3-Stage CNN stored as breast_cancer_model.keras. The project architecture describes a readable convolutional sequence rather than a large ensemble or model-selection framework.

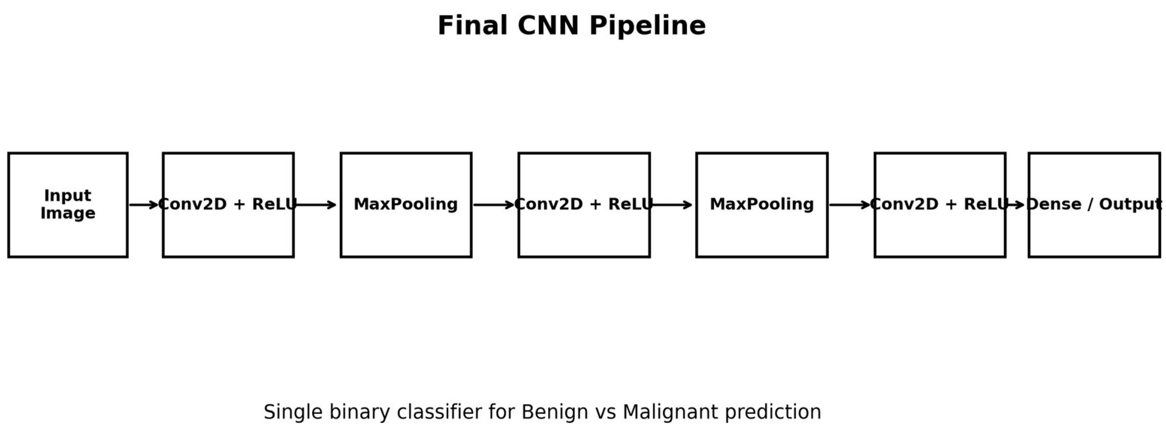

Figure 2. Simplified representation of the single CNN used for classification.

The model is used as a binary classifier for Benign and Malignant histopathology inputs. The application does not expose model-selection controls, multiple production models, or a user-adjustable malignancy threshold.

10. Training and Reproducibility

- Create patient-level train, validation, and test partitions.
- Apply preprocessing consistently.
- Apply any augmentation only to training data.
- Train the single CNN using the prepared training data.
- Monitor training and validation behavior for obvious overfitting.
- Evaluate the finalized model on the unseen test set.
- Save the trained model for later inference through the application.

The project target defined in the core requirements is a minimum of 60% test accuracy. The supplied evaluation results exceed this target at the three-class level with 81.61% accuracy.

11. Evaluation Methodology

The supplied evaluation script evaluates all images in the test directory, records the predicted class and confidence, extracts magnification and histopathological subtype from BreaKHis filenames, and computes overall class-wise, magnification-level, subtype-level, and invalid-input statistics.

Evaluation View	**Purpose**
Overall 3-class	Measures Benign, Invalid, and Malignant behavior together.
Per-class metrics	Shows precision, recall, and F1 for each class.
Confusion matrix	Shows where the model confuses classes.
Magnification analysis	Checks performance consistency at 40X, 100X, 200X, and 400X.
Subtype analysis	Shows performance across individual BreaKHis tumor subtypes.
Invalid gate test	Measures how effectively non-histopathology inputs are rejected.

12. Results

12.1 Overall Test Performance

Metric	Result
Test samples	1,664
Accuracy	81.61%
Macro Precision	80.17%
Weighted Precision	81.03%
Macro Recall	80.86%
Weighted Recall	81.61%
Macro F1	80.21%
Weighted F1	81.07%

12.2 Per-Class Performance

Class	Support	Precision	Recall	F1
Benign	486	72.89%	60.29%	65.99%
Malignant	1,103	84.48%	90.30%	87.29%
Invalid	75	83.13%	92.00%	87.34%

The most important class for the intended cancer-classification task is Malignant. The reported 90.30% malignant recall means the model correctly identified 996 of 1,103 malignant test images, while malignant precision was 84.48%. At the same time, benign performance is weaker than malignant performance, which is an important limitation and should be discussed during the exhibition rather than hidden.

12.3 Confusion Matrix

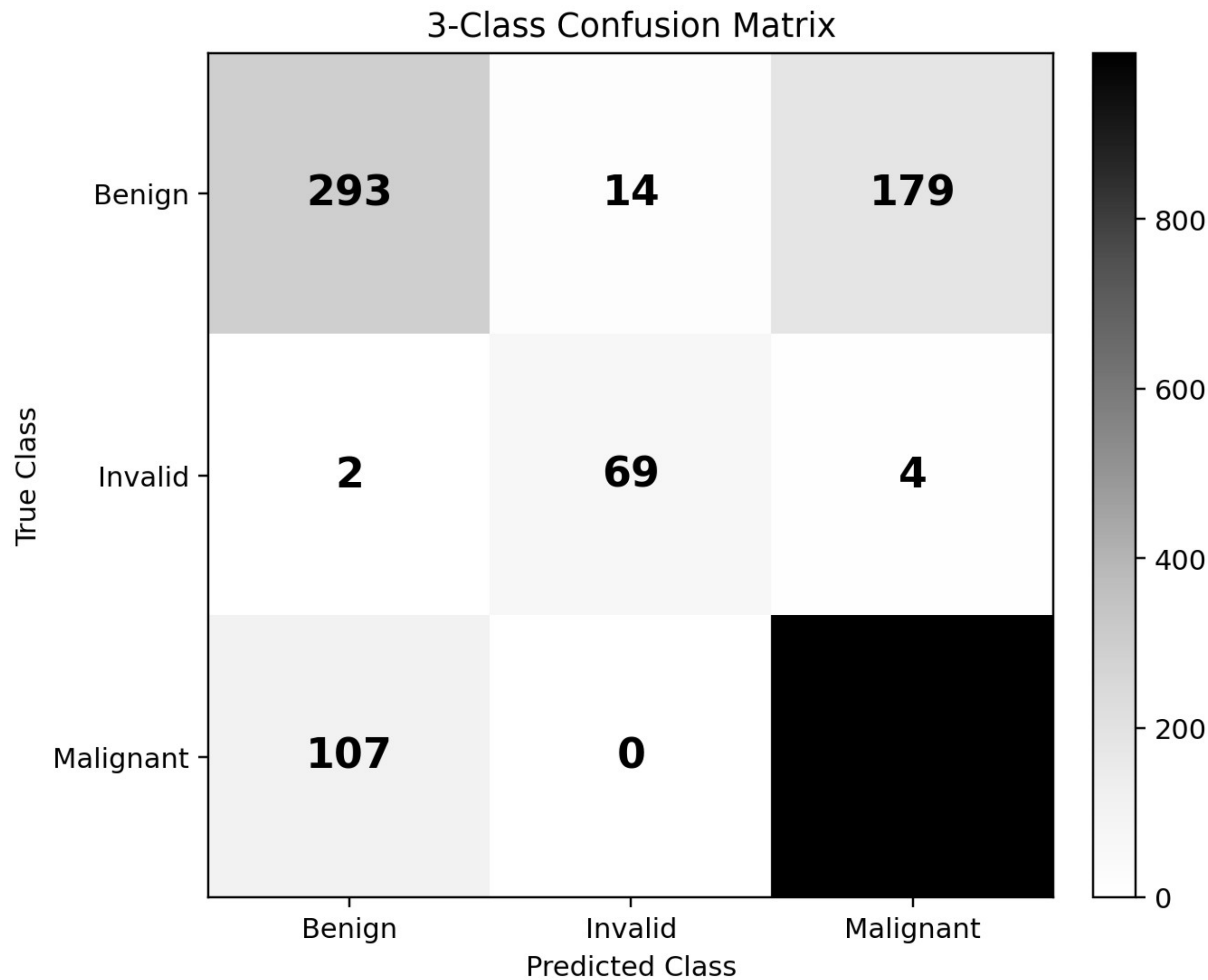

Figure 3. Confusion matrix derived from the supplied evaluation report.

The confusion matrix shows 293 correctly predicted Benign cases, 69 correctly rejected Invalid cases, and 996 correctly predicted Malignant cases. The largest cross-class error is 179 true Benign images predicted as Malignant, followed by 107 true Malignant images predicted as Benign.

12.4 Magnification-Level Performance

Magnification	Samples	Accuracy	Weighted F1
40X	392	80.87%	80.42%
100X	422	79.38%	79.85%
200X	389	81.49%	81.21%
400X	386	82.90%	82.26%

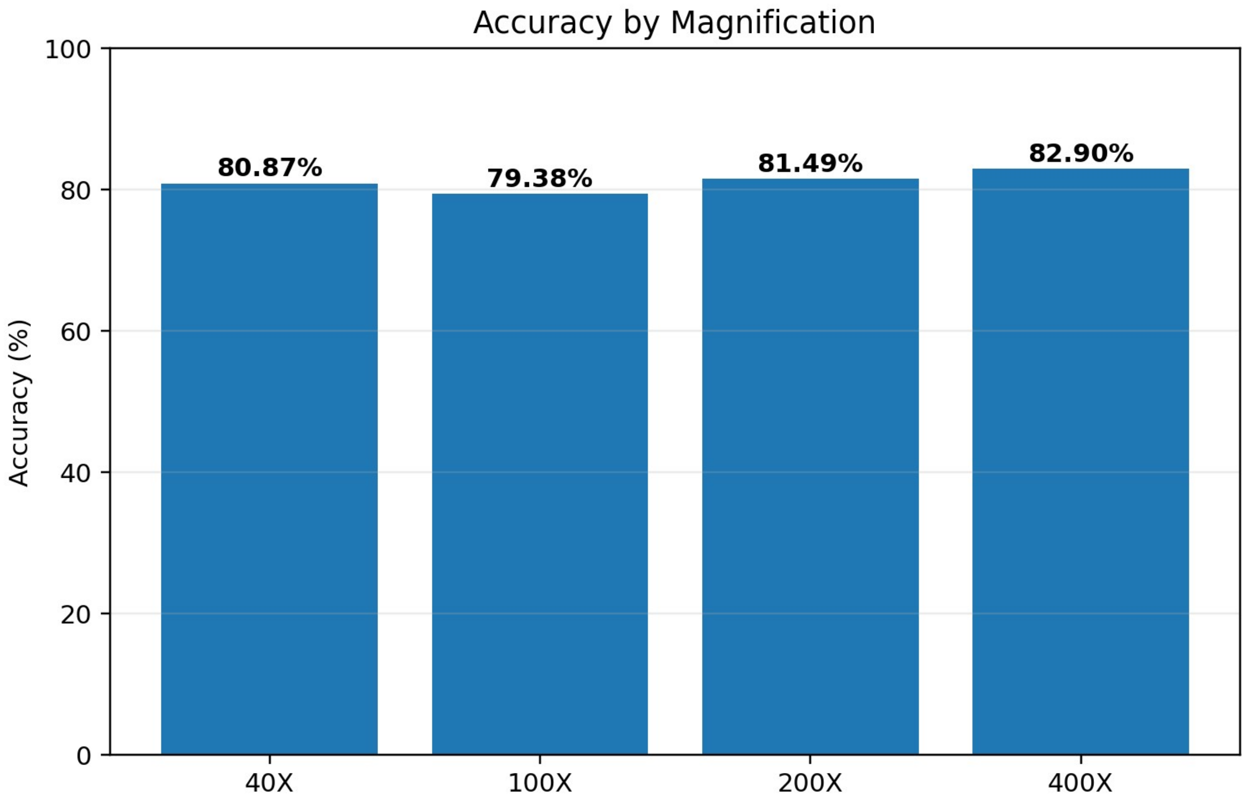

Figure 4. Accuracy remains broadly consistent across all four magnification levels.

The accuracy range across magnifications is only about 3.52 percentage points, from 79.38% at 100X to 82.90% at 400X. This supports the report claim that the model is reasonably stable across the magnification levels represented in the test set.

12.5 Histopathological Subtype Performance

Subtype	Type	Samples	Correct	Rate
Ductal Carcinoma	Malignant	258	258	100.00%
Fibroadenoma	Benign	68	0	0.00%
Lobular Carcinoma	Malignant	125	125	100.00%
Mucinous Carcinoma	Malignant	361	280	77.56%
Papillary Carcinoma	Malignant	359	333	92.76%
Phyllodes Tumor	Benign	235	124	52.77%
Tubular Adenoma	Benign	183	169	92.35%

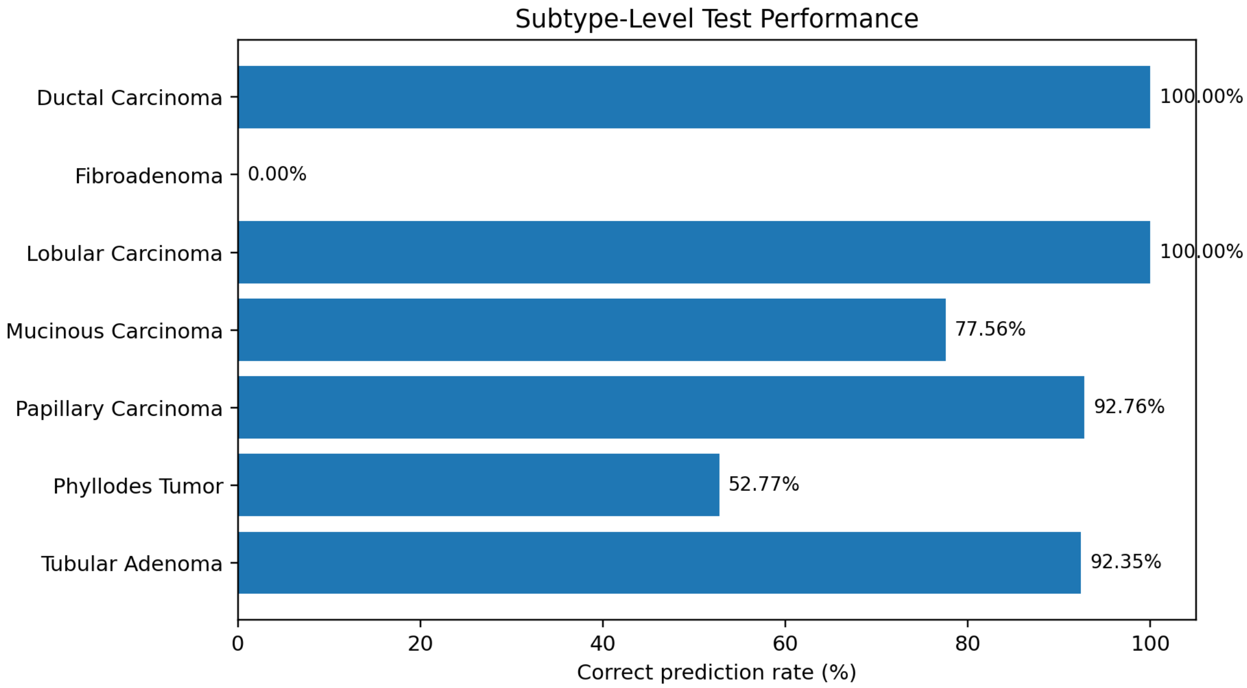

Figure 5. Subtype-level prediction rates from the supplied evaluation report.

Subtype performance reveals an important pattern: the model is very strong on several malignant subtypes but struggles substantially on Fibroadenoma and, to a lesser extent, Phyllodes Tumor. This suggests that class-level accuracy alone is not enough to understand the model; subtype-level analysis exposes where the learned representation is weaker.

12.6 Image Validation Gate

Gate Test	Result
Invalid test samples	75
Rejected correctly	69 / 75
Invalid recall	92.00%
Invalid precision	83.13%
Corrupted / invalid file handling	Passed in supplied evaluation notes

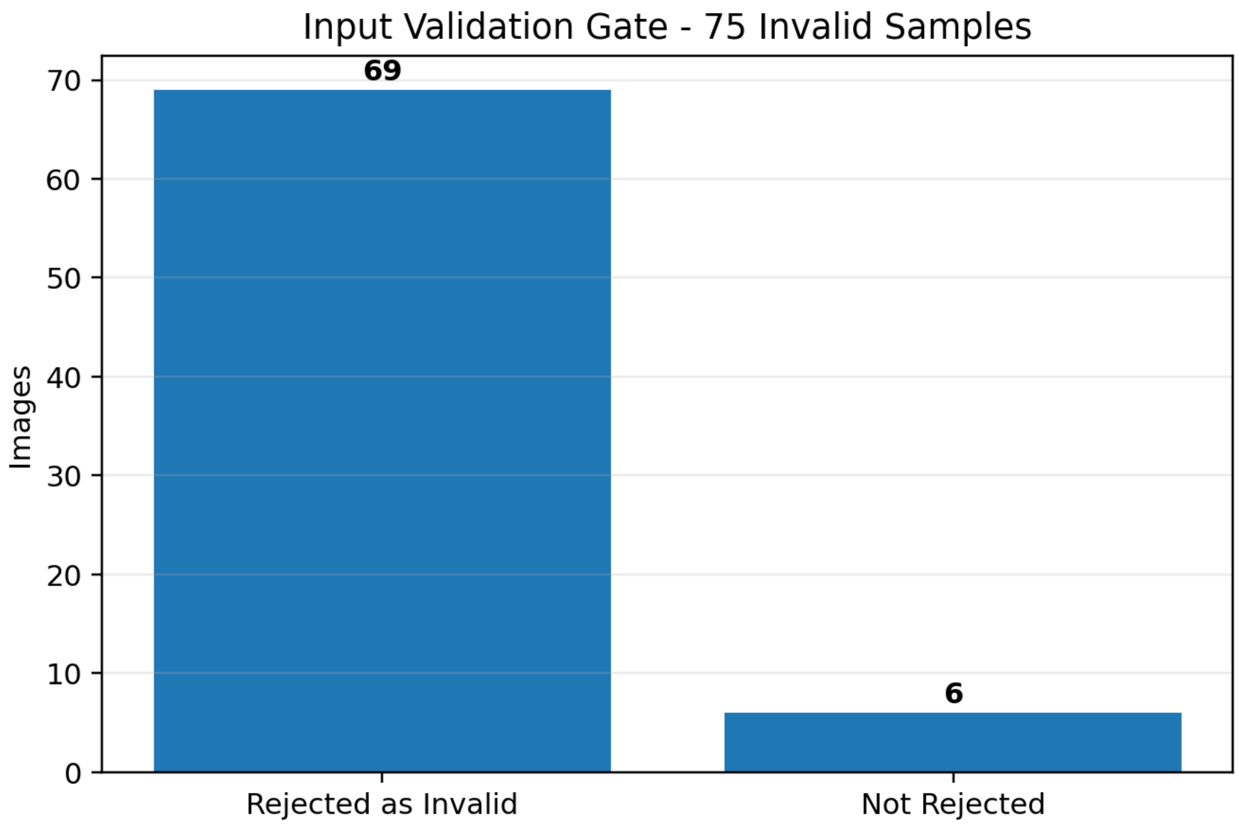

Figure 6. Input-gate performance on 75 invalid samples.

The validation gate successfully stops most invalid inputs before they can be treated as cancer-classification candidates. Six invalid examples were not rejected as invalid, which is an explicit area for further strengthening.

13. Evaluation Quality Note

The supplied MODEL_EVALUATION_REPORT.md and evaluate_detailed.py contain a small internal inconsistency in the final narrative summary. The computed overall metrics, class table, and confusion matrix support 81.61% three-class accuracy, 84.48% malignant precision, and 90.30% malignant recall. However, two later hard-coded summary lines state 74.10% accuracy and 84.04% / 85.95% malignant precision/recall. Those values are not consistent with the preceding computed table. This report therefore uses the computed metrics supported by the confusion matrix and per-class table and flags the stale hard-coded summary as a documentation issue to correct before final publication.

14. Application Design

The application layer is deliberately small. Streamlit is used as the user interface so the project can be demonstrated locally without a separate web backend. The interface follows the same logical sequence as the ML pipeline.

Screen / Step	Behavior
Upload	User selects a single image.
Validation	Image is checked before inference.
Preview	Accepted image can be shown to the user.
Prediction	Saved CNN returns Benign or Malignant.
Confidence	Model output confidence is displayed.
Safety notice	Educational/research disclaimer remains visible.

15. Testing Strategy

1. Valid histopathology image test - confirm normal prediction flow.

2. Unrelated image test - confirm validation gate rejects clearly unsuitable input.

3. Corrupted image test - confirm graceful failure.

4. Unsupported file test - confirm clear user feedback.

5. Very small or unusable image test - confirm rejection before inference.

6. Missing model test - confirm the application fails clearly rather than silently producing a result.

7. Result display test - confirm prediction, confidence, and disclaimer are visible.

16. Limitations

- The system is trained for histopathology images and should not be interpreted as a general breast-imaging system.
- Benign performance is materially weaker than malignant performance in the supplied test results.
- Subtype behavior is uneven, with especially weak performance on Fibroadenoma in the reported test set.
- The image-validation gate rejects most, but not all, invalid inputs in the supplied test set.
- Model confidence is not a clinically calibrated probability of disease.
- The supplied report contains stale hard-coded summary values that should be corrected before external publication.

17. Future Improvements

- Improve the invalid-image gate using a larger and more representative negative-image test set.
- Investigate why benign subtypes, particularly Fibroadenoma and Phyllodes Tumor, are confused with malignant images.
- Use class-aware analysis to improve balanced performance rather than optimizing only one aggregate metric.

- Evaluate additional preprocessing or augmentation strategies based on observed failure modes.
- Add more robust reporting automation so narrative summaries are generated directly from computed metrics rather than hard-coded values.
- Consider additional modalities only in a later project phase after the histopathology pipeline is stable and individually validated.

18. Reproducibility and Repository Structure

The project is intended to remain reproducible and understandable when moved into a version-controlled repository. The repository should prioritize source code, requirements, application files, and project reports while keeping very large datasets outside ordinary source control when appropriate.

Path	**Purpose**
app.py	Streamlit application
train.py	Model training pipeline
predict.py	Model loading and inference
preprocess.py	Shared preprocessing logic
model.py	CNN definition
requirements.txt	Python dependencies
data/	Local dataset location; large raw datasets may be external to the repository
models/	Saved trained model artifacts
reports/	Project and evaluation reports

19. Responsible Use and Medical Positioning

This system must be presented as an AI/ML educational and research prototype. It is not a medical device and does not replace a pathologist, radiologist, clinician, or other qualified medical professional. Prediction confidence represents the model output and should not be interpreted as a guaranteed medical probability.

20. Team and Contributions

Role	**Name**
Author / Project Lead	Shwetank Vaibhav
Contributor	Pranshu Gupta
Contributor	Ayush Shukla
Contributor	Debasish Kumar Sahoo
Contributor	Manvendra Kumar
Contributor	Piyush Kumar Das

The project is a collaborative academic build spanning research, machine learning, software development, testing, documentation, and exhibition preparation.

21. Conclusion

Project Exhibition 01 demonstrates a complete, compact AI/ML pipeline for breast histopathology image classification. Its strongest design feature is the clear separation between image validation and cancer prediction: an uploaded image is first checked for suitability and only then passed into the classifier. This keeps the system safer, easier to explain, and more faithful to the intended scope.

The supplied results show that the final 3-class evaluation reaches 81.61% accuracy on 1,664 unseen test images, comfortably above the project's initial 60% target. Malignant recall reaches 90.30%, and the invalid-input gate achieves 92.00% recall. At the same time, the results reveal meaningful limitations - particularly weaker benign subtype performance - demonstrating why a good project report must discuss failure cases as carefully as successes.

Overall, the project succeeds as a student-focused demonstration of dataset preparation, controlled image validation, CNN-based classification, quantitative evaluation, and application deployment. The next level of improvement should come from evidence-driven refinement of the weak cases rather than from simply adding more complexity.

Appendix A. Key Commands

```
python -m venv venv
```

```
pip install -r requirements.txt
```

```
streamlit run app.py
```

Appendix B. Primary Evaluation Snapshot

Test set: 1,664 images | Overall accuracy: 81.61% | Weighted precision: 81.03% | Weighted recall: 81.61% | Weighted F1: 81.07% | Invalid recall: 92.00%